Constituent (linguistics)

From Wikipedia, the free encyclopedia

In syntactic analysis, a **constituent** is a word or a group of words that function as a single unit within a hierarchical structure. The constituent structure of sentences is identified using *tests for constituents*.[1] These tests apply to a portion of a sentence, and the results provide evidence about the constituent structure of the sentence. Many constituents are phrases. A phrase is a sequence of one or more words (in some theories two or more) built around a head lexical item and working as a unit within a sentence. A word sequence is shown to be a phrase/constituent if it exhibits one or more of the behaviors discussed below. The analysis of constituent structure is associated mainly with phrase structure grammars, although dependency grammars also allow sentence structure to be broken down into constituent parts.

Tests for constituents in English [edit]

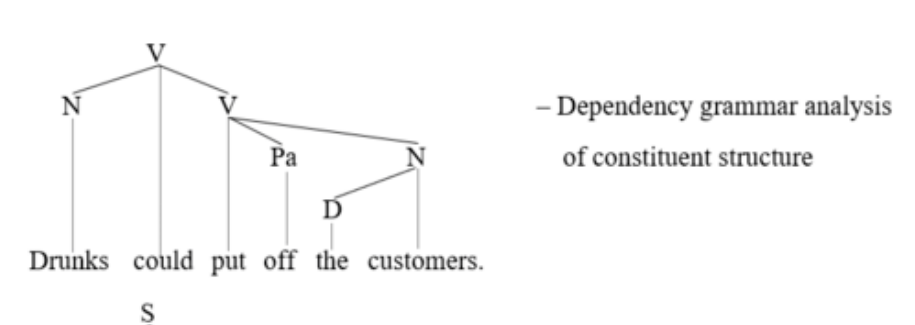
Tests for constituents are diagnostics used to identify sentence structure. There are numerous tests for constituents that are commonly used to identify the constituents of English sentences. 15 of the most commonly used tests are listed next: 1) coordination (conjunction), 2) pro-form substitution (replacement), 3) topicalization (fronting), 4) *do-so*-substitution, 5) *one*-substitution, 6) answer ellipsis (question test), 7) clefting, 8) VP-ellipsis, 9) pseudoclefting, 10) passivization, 11) omission (deletion), 12) intrusion, 13) wh-fronting, 14) general substitution, 15) right node raising (RNR).

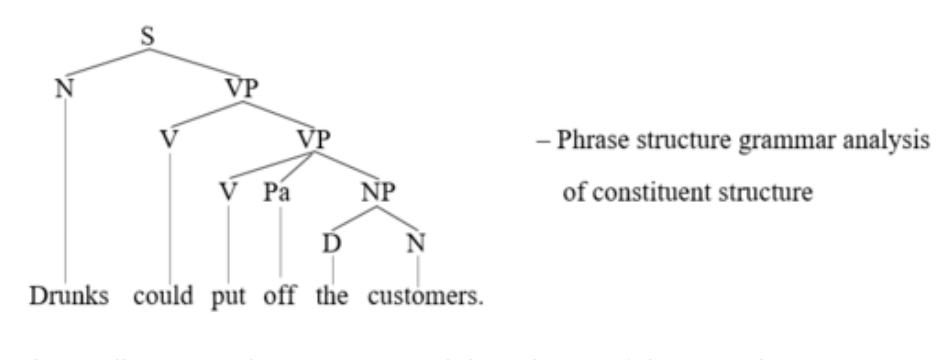
The order in which these 15 tests are listed here corresponds to the frequency of use, coordination being the most frequently used of the 15 tests and RNR being the least frequently used. A general word of caution is warranted when employing these tests, since they often deliver contradictory results. The tests are merely rough-and-ready tools that grammarians employ to reveal clues about syntactic structure. Some syntacticians even arrange the tests on a scale of reliability, with less-reliable tests treated as useful to confirm constituency though not sufficient on their own. Failing to pass a single test does not mean that the test string is not a constituent, and conversely, passing a single test does not necessarily mean the test string is a constituent. It is best to apply as many tests as possible to a given string in order to prove or to rule out its status as a constituent.

The 15 tests are introduced, discussed, and illustrated below mainly relying on the same one sentence:[2]

Drunks could put off the customers.

By restricting the introduction and discussion of the tests for constituents below mainly to this one sentence, it becomes possible to compare the results of the tests. To aid the discussion and illustrations of the constituent structure of this sentence, the following two sentence diagrams are employed (D = determiner, N = noun, NP = noun phrase, Pa = particle, S = sentence, V = Verb, VP = verb phrase):


V
N
V
Pa
N
D
Drunks could put off the customers.
– Dependency grammar analysis of constituent structure


S
N
VP
V
VP
V
Pa
NP
D
N
Drunks could put off the customers.
– Phrase structure grammar analysis of constituent structure

These diagrams show two potential analyses of the constituent structure of the sentence. A given node in a tree diagram is understood as marking a constituent, that is, a constituent is understood as corresponding to a given node and everything that that node exhaustively dominates. Hence the first tree, which shows the constituent structure according to dependency grammar, marks the following words and word combinations as constituents: *Drunks*, *off*, *the*, *the customers*, and *put off the customers*.[3] The second tree, which shows the constituent structure according to phrase structure grammar, marks the following words and word combinations as constituents: *Drunks*, *could*, *put*, *off*, *the*, *customers*, *the customers*, *put off the customers*, and *could put off the customers*. The analyses in these two tree diagrams provide orientation for the discussion of tests for constituents that now follows.

Coordination [edit]

The coordination test assumes that only constituents can be coordinated, i.e., joined by means of a coordinator such as *and*, *or*, or *but*:[4] The next examples demonstrate that coordination identifies individual words as constituents:

Drunks could put off the customers.
(a) [**Drunks**] and [bums] could put off the customers.
(b) Drunks [**could**] and [would] put off the customers.
(c) Drunks could [**put off**] and [drive away] the customers.
(d) Drunks could put off the [**customers**] and [neighbors].

The square brackets mark the conjuncts of the coordinate structures. Based on these data, one might assume that *drunks*, *could*, *put off*, and *customers* are constituents in the test sentence because these strings can be coordinated with *bums*, *would*, *drive away*, and *neighbors*, respectively. Coordination also identifies multi-word strings as constituents:

(e) Drunks could put off [**the customers**] and [the neighbors].
(f) Drunks could [**put off the customers**] and [drive away the neighbors].
(g) Drunks [**could put off the customers**] and [would drive away the neighbors].

These data suggest that *the customers*, *put off the customers*, and *could put off the customers* are constituents in the test sentence.

Examples such as (a-g) are not controversial insofar as many theories of sentence structure readily view the strings tested in sentences (a-g) as constituents. However, additional data are problematic, since they suggest that certain strings are also constituents even though most theories of syntax do not acknowledge them as such, e.g.

(h) Drunks [**could put off**] and [would really annoy] the customers.
(i) Drunks could [**put off these**] and [piss off those] customers.
(j) [**Drunks could**], and [they probably would], put off the customers.

These data suggest that *could put off*, *put off these*, and *Drunks could* are constituents in the test sentence. Most theories of syntax reject the notion that these strings are constituents, though. Data such as (h-j) are sometimes addressed in terms of the right node raising (RNR) mechanism.

The problem for the coordination test represented by examples (h-j) is compounded when one looks beyond the test sentence, for one quickly finds that coordination suggests that a wide range of strings are constituents that most theories of syntax do not acknowledge as such, e.g.

(k) Sam leaves [**from home on Tuesday**] and [from work on Wednesday].
(l) Sam leaves [**from home on Tuesday on his bicycle**] and [from work on Wednesday in his car].
(m) Sam leaves [**from home on Tuesday**], and [from work].

The strings *from home on Tuesday* and *from home on Tuesday on his bicycle* are not viewed as constituents in most theories of syntax, and concerning sentence (m), it is very difficult there to even discern how one should delimit the conjuncts of the coordinate structure. The coordinate structures in (k-l) are sometimes characterized in terms of non-constituent conjuncts (NCC), and the instance of coordination in sentence (m) is sometimes discussed in terms of stripping and/or gapping.

Due to the difficulties suggested with examples (h-m), many grammarians view coordination skeptically regarding its value as a test for constituents. The discussion of the other tests for constituents below reveals that this skepticism is warranted, since coordination identifies many more strings as constituents than the other tests for constituents.[5]

Proform substitution (replacement) [edit]

Proform substitution, or replacement, involves replacing the test string with the appropriate proform (e.g. pronoun, pro-verb, pro-adjective, etc.). Substitution normally involves using a definite proform like *it*, *he*, *there*, *here*, etc. in place of a phrase or a clause. If such a change yields a grammatical sentence where the general structure has not been altered, then the test string is likely a constituent:[6]

Drunks could put off the customers.
(a) **They** could put off the customers. (*They* = ***Drunks***)
(b) Drunks could put **them** off. (*them* = ***the customers***; note that shifting of *them* and *off* has occurred here.)
(c) Drunks could **do it**. (*do it* = ***put off the customers***)

These examples suggest that *Drunks*, *the customers*, and *put off the customers* in the test sentence are constituents. An important aspect of the proform test is the fact that it fails to identify most subphrasal strings as constituents, e.g.

(d) *Drunks **do so/it** put off the customers (*do so/it* = *could*)
(e) *Drunks could **do so/it** off the customers (*do so/it* = *put*)
(f) *Drunks could put **so/it** the customers (*so/it* = *off*)
(g) *Drunks could put off the **them**. (*them* = *customers*)

These examples suggest that the individual words *could*, *put*, *off*, and *customers* should not be viewed as constituents. This suggestion is of course controversial, since most theories of syntax assume that individual words are constituents by default. The conclusion one can reach based on such examples, however, is that proform substitution using a definite proform identifies phrasal constituents only; it fails to identify sub-phrasal strings as constituents.

Topicalization (fronting) [edit]

Topicalization involves moving the test string to the front of the sentence. It is a simple movement operation.[7] Many instances of topicalization seem only marginally acceptable when taken out of context. Hence to suggest a context, an instance of topicalization can be preceded by *...and* and a modal adverb can be added as well (e.g. *certainly*):

Drunks could put off the customers.
(a) ...and **the customers**, drunks certainly could put off.
(b) ...and **put off the customers**, drunks certainly could.

These examples suggest that *the customers* and *put off the customers* are constituents in the test sentence. Topicalization is like many of the other tests in that it identifies phrasal constituents only. When the test sequence is a sub-phrasal string, topicalization fails:

(c) *...and **customers**, drunks certainly could put off the.
(d) *...and **could**, drunks certainly put off the customers.
(e) *...and **put**, drunks certainly could off the customers.
(f) *...and **off**, drunks certainly could put the customers.
(g) *...and **the**, drunks certainly could put off customers.

These examples demonstrate that *customers*, *could*, *put*, *off*, and *the* fail the topicalization test. Since these strings are all sub-phrasal, one can conclude that topicalization is unable to identify sub-phrasal strings as constituents.

Do-so-substitution [edit]

Do-so-substitution is a test that substitutes a form of *do so* (*does so*, *did so*, *done so*, *doing so*) into the test sentence for the target string. This test is widely used to probe the structure of strings containing verbs (because *do* is a verb).[8] The test is limited in its applicability, though, precisely because it is only applicable to strings containing verbs:

Drunks could put off the customers.
(a) Drunks could **do so**. (*do so* = ***put off the customers***)
(b) Drunks **do so**. (*do so* ≠ ***could put off the customers***)

The 'a' example suggests that *put off the customers* is a constituent in the test sentence, whereas the b example fails to suggest that *could put off the customers* is a constituent, for *do so* cannot include the meaning of the modal verb *could*. To illustrate more completely how the *do so* test is employed, another test sentence is now used, one that contains two post-verbal adjunct phrases:

We met them in the pub because we had time.
(c) We **did so** in the pub because we had time. (*did so* = ***met them***)
(d) We **did so** because we had time. (*did so* = ***met them in the pub***)
(e) We **did so**. (*did so* = ***met them in the pub because we had time***)

These data suggest that *met them*, *met them in the pub*, and *met them in the pub because we had time* are constituents in the test sentence. Taken together, such examples seem to motivate a structure for the test sentence that has a left-branching verb phrase, because only a left-branching verb phrase can view each of the indicated strings as a constituent. There is a problem with this sort of reasoning, however, as the next example illustrates:

(f) We **did so** in the pub. (*did so* = ***met them because we had time***)

In this case, *did so* appears to stand in for the discontinuous word combination consisting of *met them* and *because we had time*. Such a discontinuous combination of words cannot be construed as a constituent. That such an interpretation of *did so* is indeed possible is seen in a fuller sentence such as *You met them in the cafe because you had time, and we did so in the pub*. In this case, the preferred reading of *did so* is that it indeed simultaneously stands in for both *met them* and *because we had time*.

One-substitution [edit]

The *one*-substitution test replaces the test string with the indefinite pronoun *one* or *ones*.[9] If the result is acceptable, then the test string is deemed a constituent. Since *one* is a type of pronoun, *one*-substitution is only of value when probing the structure of noun phrases. In this regard, the test sentence from above is expanded in order to better illustrate the manner in which one-substitution is generally employed:

Drunks could put off the loyal customers around here who we rely on.
(a) Drunks could put off the loyal **ones** around here who we rely on. (ones = ***customers***)
(b) Drunks could put off the **ones** around here who we rely on. (*ones* = ***loyal customers***)
(c) Drunks could put off the loyal **ones** who we rely on. (*ones* = ***customers around here***)
(d) Drunks could put off the **ones** who we rely on. (*ones* = ***loyal customers around here***)
(e) Drunks could put off the loyal **ones**. (*ones* = ***customers around here who we rely on***)

These examples suggest that *customers*, *loyal customers*, *customers around here*, *loyal customers around here*, and *customers around here who we rely on* are constituents in the test sentence. Some have pointed to a problem associated with the *one*-substitution in this area, however. This problem is that it is impossible to produce a single constituent structure of the noun phrase *the loyal customers around here who we rely on* that could simultaneous view all of the indicated strings as constituents.[10] Another problem that has been pointed out concerning the *one*-substitution as a test for constituents is the fact that it at times suggests that non-string word combinations are constituents,[11] e.g.

(f) Drunks would put off the **ones** around here. (ones = **loyal customers who we rely on**)

The word combination consisting of both *loyal customers* and *who we rely on* is discontinuous in the test sentence, a fact that should motivate one to generally question the value of *one*-substitution as a test for constituents.

Answer fragments (answer ellipsis, question test, standalone test) [edit]

The answer fragment test involves forming a question that contains a single wh-word (e.g. *who*, *what*, *where*, etc.). If the test string can then appear alone as the answer to such a question, then it is likely a constituent in the test sentence:[12]

Drunks could put off the customers.
(a) Who could put off the customers? - **Drunks**.
(b) Who could drunks put off? - **The customers**.
(c) What would drunks do? - **Put off the customers**.

These examples suggest that *Drunks*, *the customers*, and *put off the customers* are constituents in the test sentence. The answer fragment test is like most of the other tests for constituents in that it does not identify sub-phrasal strings as constituents:

(d) What about putting off the customers? - ***Could**.
(e) What could drunks do about the customers? - ***Put**.
(f) *What could drunks do about putting the customers? - ***Off**.
(g) *Who could drunks put off the? - ***Customers**.

These answer fragments are all grammatically unacceptable, suggesting that *could*, *put*, *off*, and *customers* are not constituents. Note as well that the latter two questions themselves are ungrammatical. It is apparently often impossible to form the question in a way that could successfully elicit the indicated strings as answer fragments. The conclusion, then, is that the answer fragment test is like most of the other tests in that it fails to identify sub-phrasal strings as constituents.

Clefting [edit]

Clefting involves placing the test string X within the structure beginning with *It is/was*: *It was X that...*.[13] The test string appears as the pivot of the cleft sentence:

Drunks could put off the customers.
(a) It is **drunks** that could put off the customers.
(b) It is **the customers** that drunks could put off.
(c) ??It is **put off the customers** that drunks could do.

These examples suggest that *Drunks* and *the customers* are constituents in the test sentence. Example c is of dubious acceptability, suggesting that *put off the customers* may not be constituent in the test string. Clefting is like most of the other tests for constituents in that it fails to identify most individual words as constituents:

(d) *It is **could** that drunks put off the customers.
(e) *It is **put** that drunks could off the customers.
(f) *It is **off** that drunks could put the customers.
(g) *It is **the** that drunks could put off customers.
(h) *It is **customers** that drunks could put off the.

The examples suggest that each of the individual words *could*, *put*, *off*, *the*, and *customers* are not constituents, contrary to what most theories of syntax assume. In this respect, clefting is like many of the other tests for constituents in that it only succeeds at identifying certain phrasal strings as constituents.

VP-ellipsis (verb phrase ellipsis) [edit]

The VP-ellipsis test checks to see which strings containing one or more predicative elements (usually verbs) can be elided from a sentence. Strings that can be elided are deemed constituents:[14] The symbol ∅ is used in the following examples to mark the position of ellipsis:

Beggars could immediately put off the customers when they arrive, and
(a) *drunks could immediately also ∅ the customers when they arrive. (∅ = **put off**)
(b) ?drunks could immediately also ∅ when they arrive. (∅ = **put off the customers**)
(c) drunks could also ∅ when they arrive. (∅ = **immediately put off the customers**)
(d) drunks could immediately also ∅. (∅ = **put off the customers when they arrive**)
(e) drunks could also ∅. (∅ = **immediately put off the customers when they arrive**)

These examples suggest that *put off* is not a constituent in the test sentence, but that *immediately put off the customers*, *put off the customers when they arrive*, and *immediately put off the customers when they arrive* are constituents. Concerning the string *put off the customers* in (b), marginal acceptability makes it difficult to draw a conclusion about *put off the customers*.

There are various difficulties associated with this test. The first of these is that it can identify too many constituents, such as in this case here where it is impossible to produce a single constituent structure that could simultaneously view each of the three acceptable examples (c-e) as having elided a constituent. Another problem is that the test can at times suggest that a discontinuous word combination is a constituent, e.g.:

(f) Frank will help tomorrow in the office, and Susan will ∅ today. (∅ = **help...in the office**)

In this case, it appears as though the elided material corresponds to the discontinuous word combination including *help* and *in the office*.

Pseudoclefting [edit]

Pseudoclefting is similar to clefting in that it puts emphasis on a certain phrase in a sentence. There are two variants of the pseudocleft test. One variant inserts the test string X in a sentence starting with a free relative clause: *What.....is/are X*; the other variant inserts X at the start of the sentence followed by the *it/are* and then the free relative clause: *X is/are what/who...* Only the latter of these two variants is illustrated here.[15]

Drunks would put off the customers.
(a) **Drunks** are who could put off the customers.
(b) **The customers** are who drunks could put off.
(c) **Put off the customers** is what drunks could do.

These examples suggest that *Drunks*, *the customers*, and *put off the customers* are constituents in the test sentence. Pseudoclefting fails to identify most individual words as constituents:

(d) ***Could** is what drunks put off the customers.
(e) ***Put** is what drunks could off the customers.
(f) ***Off** is what drunks could put the customers.
(g) ***The** is who drunks could put off customers.
(h) ***Customers** is who drunks could put off the.

The pseudoclefting test is hence like most of the other tests insofar as it identifies phrasal strings as constituents, but does not suggest that sub-phrasal strings are constituents.

Passivization [edit]

Passivization involves changing an active sentence to a passive sentence, or vice versa. The object of the active sentence is changed to the subject of the corresponding passive sentence:[16]

(a) **Drunks** could put off **the customers**.
(b) **The customers** could be put off by **drunks**.

The fact that sentence (b), the passive sentence, is acceptable, suggests that *Drunks* and *the customers* are constituents in sentence (a). The passivization test used in this manner is only capable of identifying subject and object words, phrases, and clauses as constituents. It does not help identify other phrasal or sub-phrasal strings as constituents. In this respect, the value of passivization as test for constituents is very limited.

Omission (deletion) [edit]

Omission checks whether the target string can be omitted without influencing the grammaticality of the sentence. In most cases, local and temporal adverbials, attributive modifiers, and optional complements can be safely omitted and thus qualify as constituents.[17]

Drunks could put off the customers.
(a) Drunks could put off customers. (***the*** has been omitted.)

This sentence suggests that the definite article *the* is a constituent in the test sentence. Regarding the test sentence, however, the omission test is very limited in its ability to identify constituents, since the strings that one wants to check do not appear optionally. Therefore, the test sentence is adapted to better illustrate the omission test:

The **obnoxious** drunks could **immediately** put off the customers **when they arrive**.
(b) The drunks could immediately put off the customers when they arrive. (***obnoxious*** has been successfully omitted.)
(c) The obnoxious drunks could put off the customers when they arrive. (***immediately*** has been successfully omitted.)
(d) The obnoxious drunks could immediately put off the customers. (***when they arrive*** has been successfully omitted.)

The ability to omit *obnoxious*, *immediately*, and *when they arrive* suggests that these strings are constituents in the test sentence. Omission used in this manner is of limited applicability, since it is incapable of identifying any constituent that appears obligatorily. Hence there are many target strings that most accounts of sentence structure take to be constituents but that fail the omission test because these constituents appear obligatorily, such as subject phrases.

Intrusion [edit]

Intrusion probes sentence structure by having an adverb "intrude" into parts of the sentence. The idea is that the strings on either side of the adverb are constituents.[18]

Drunks could put off the customers.
(a) Drunks **definitely** could put off the customers.
(b) Drunks could **definitely** put off the customers.
(c) *Drunks could put **definitely** off the customers.
(d) *Drunks could put off **definitely** the customers.
(e) *Drunks could put off the **definitely** customers.

Example (a) suggests that *Drunks* and *could put off the customers* are constituents. Example (b) suggests that *Drunks could* and *put off the customers* are constituents. The combination of (a) and (b) suggest in addition that *could* is a constituent. Sentence (c) suggests that *Drunks could put* and *off the customers* are not constituents. Example (d) suggests that *Drunks could put off* and *the customers* are not constituents. And example (e) suggests that *Drunks could put off the* and *customers* are not constituents.

Those that employ the intrusion test usually use a modal adverb like *definitely*. This aspect of the test is problematic, though, since the results of the test can vary based upon the choice of adverb. For instance, manner adverbs distribute differently than modal adverbs and will hence suggest a distinct constituent structure from that suggested by modal adverbs.

Wh-fronting [edit]

Wh-fronting checks to see if the test string can be fronted as a wh-word.[19] This test is similar to the answer fragment test insofar it employs just the first half of that test, disregarding the potential answer to the question.

Drunks would put off the customers.
(a) **Who** would put off the customers? (*Who* ↔ ***Drunks***)
(b) **Who** would drunks put off? (*Who* ↔ ***the customers***)
(c) **What** would drunks do? (*What...do* ↔ ***put off the customers***)

These examples suggest that *Drunks*, *the customers*, and *put off the customers* are constituents in the test sentence. Wh-fronting is like a number of the other tests in that it fails to identify many subphrasal strings as constituents:

(d) ***Do what** drunks put off the customers? (*Do what* ↔ ***would***)
(e) ***Do what** drunks would off the customers? (*Do what* ↔ ***put***)
(f) ***What** would drunks put the customers? (*What* ↔ ***off***)
(g) ***What** would drunks put off customers? (*What* ↔ ***the***)
(h) ***Who** would drunks put off the? (*Who* ↔ ***customers***)

These examples demonstrate a lack of evidence for viewing the individual words *would*, *put*, *off*, *the*, and *customers* as constituents.

General substitution [edit]

The general substitution test replaces the test string with some other word or phrase.[20] It is similar to proform substitution, the only difference being that the replacement word or phrase is not a proform, e.g.

Drunks could put off the customers.
(a) **Beggars** could put off the customers. (Beggars ↔ **Drunks**)
(b) Drunks could put off **our guests**. (our guests ↔ **the customers**)
(c) Drunks **would** put off the customers. (would ↔ **could**)

These examples suggest that the strings *Drunks*, *the customers*, and *could* are constituents in the test sentence. There is a major problem with this test, for it is easily possible to find a replacement word for strings that the other tests suggest are clearly not constituents, e.g.

(d) Drunks **piss** off the customers. (piss ↔ **could put**)
(e) **Beggars** put off the customers. (Beggars ↔ **Drunks could**)
(f) Drunks **like** customers. (like ↔ **could put off the**)

These examples suggest that *could put*, *Drunks could*, and *could put off the* are constituents in the test sentence. This is contrary to what the other tests reveal and to what most theories of sentence structure assume. The value of general substitution as test for constituents is therefore suspect. It is like the coordination test in that it suggests that too many strings are constituents.

Right node raising (RNR) [edit]

Right node raising, abbreviated as RNR, is a test that isolates the test string on the right side of a coordinate structure.[21] The assumption is that only constituents can be shared by the conjuncts of a coordinate structure, e.g.

Drunks could put off the customers.
(a) [Drunks] and [beggars] **could put off the customers**.
(b) [Drunks could], and [they probably would], **put off the customers**.
(c) [Drunks could approach] and [they would then put off] **the customers**.

These examples suggest that *could put off the customers*, *put off the customers*, and *the customers* are constituents in the test sentence. There are two problems with the RNR diagnostic as a test for constituents. The first is that it is limited in its applicability, since it is only capable of identifying strings as constituents if they appear on the right side of the test sentence. The second is that it can suggest strings to be constituents that most of the other tests suggest are not constituents. To illustrate this point, a different example must be used:

Frank has given his bicycle to us to use if need be.
(d) [Frank has offered], and [Susan has already loaned], **their bicycles to us to use if need be**.
(e) [Frank has offered his bicycle] and [Susan has already loaned her bicycle] **to us to use if need be**.
(f) [Frank has offered his bicycle to us] and [Susan has already loaned her bicycle to us] **to use if need be**.

These examples suggest that *their bicycles (his bicycle) to us to use if need be*, *to us to use if need be*, and *to use if need be* are constituents in the test sentence. Most theories of syntax do not view these strings as constituents, and more importantly, most of the other tests suggest that they are not constituents. In short, these tests are not taken for granted because a constituent may pass one test and fail to pass many others. We need to consult our intuitive thinking when judging the constituency of any set of words.

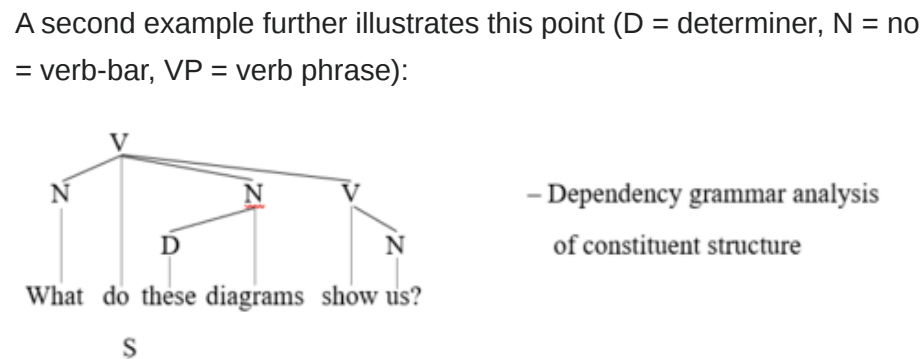
Other languages [edit]

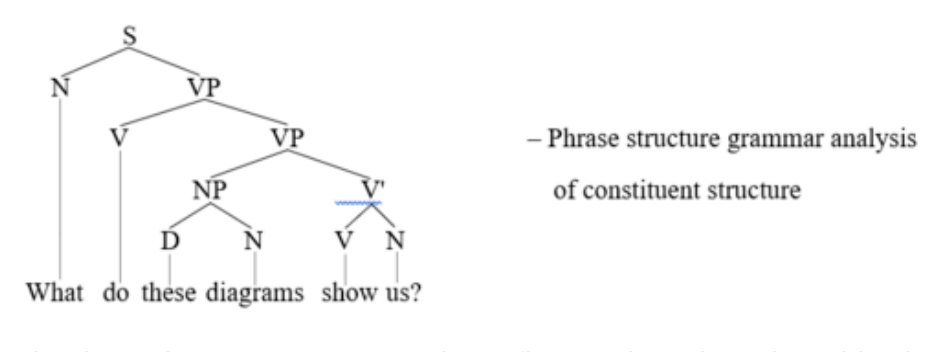
A word of caution is warranted concerning the tests for constituents as just discussed above. These tests are found in textbooks on linguistics and syntax that are written mainly with the syntax of English in mind, and the examples that are discussed are mainly from English. The tests may or may not be valid and useful when probing the constituent structure of other languages. Ideally, a battery of tests for constituents can and should be developed for each language, catered to the idiosyncrasies of the language at hand.

Competing theories [edit]

Constituent structure analyses of sentences are a central concern for theories of syntax. A given theory can produce an analysis of constituent structure that is quite unlike the next. This point is evident with the two tree diagrams above of the sentence *Drunks could put off the customers*, where the dependency grammar analysis of constituent structure looks very much unlike the phrase structure analysis. The crucial difference across the two analyses is that the phrase structure analysis views every individual word as a constituent by default, whereas the dependency grammar analysis sees only those individual words as constituents that do not dominate other words. Phrase structure grammars therefore acknowledge many more constituents than dependency grammars.

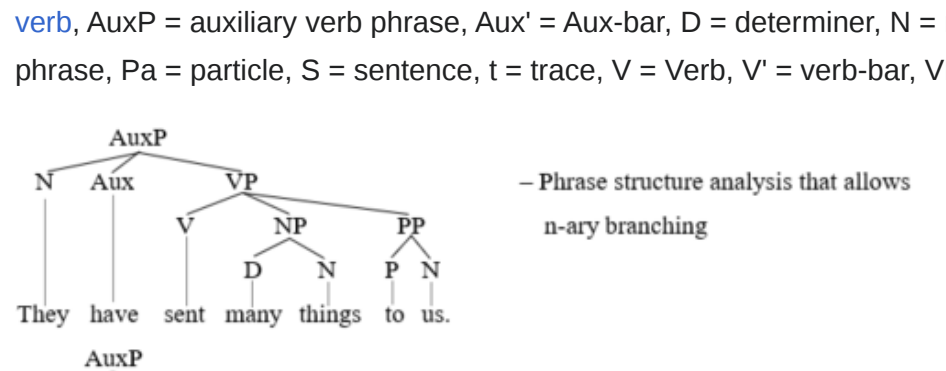
A second example further illustrates this point (D = determiner, N = noun, NP = noun phrase, Pa = particle, S = sentence, V = Verb, V' = verb-bar, VP = verb phrase):

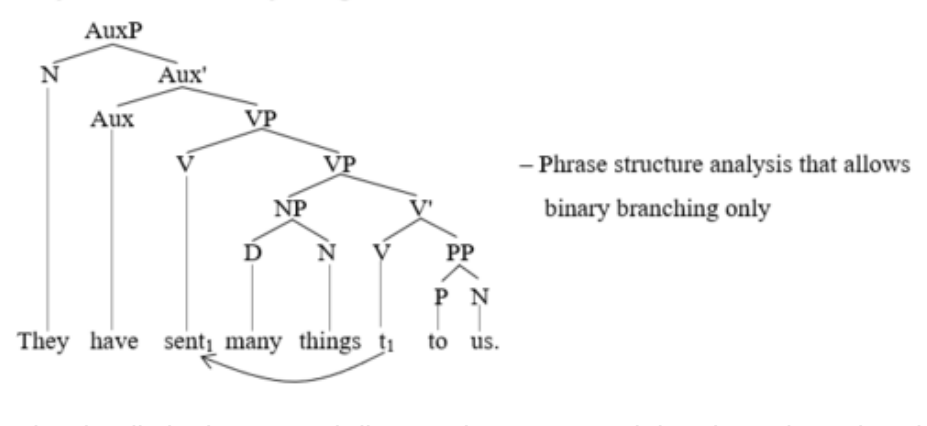

V
N
N
V
D
N
What do these diagrams show us?
– Dependency grammar analysis of constituent structure


S
N
VP
V
VP
NP
V'
D
N
V
N
What do these diagrams show us?
– Phrase structure grammar analysis of constituent structure

The dependency grammar tree shows five words and word combinations as constituents: *who*, *these*, *us*, *these diagrams*, and *show us*. The phrase structure tree, in contrast, shows nine words and word combinations as constituents: *what*, *do*, *these*, *diagrams*, *show*, *us*, *these diagrams*, *show us*, and *do these diagrams show us*. The two diagrams thus disagree concerning the status of *do*, *diagrams*, *show*, and *do these diagrams show us*, the phrase structure diagram showing them as constituents and the dependency grammar diagram showing them as non-constituents. To determine which analysis is more plausible, one turns to the tests for constituents discussed above.[22]

Within phrase structure grammars, views about of constituent structure can also vary significantly. Many modern phrase structure grammars assume that syntactic branching is always binary, that is, each greater constituent is necessarily broken down into two lesser constituents. More dated phrase structures analyses are, however, more likely to allow n-ary branching, that is, each greater constituent can be broken down into one, two, or more lesser constituents. The next two trees illustrate the distinction (Aux = auxiliary verb, AuxP = auxiliary verb phrase, Aux' = Aux-bar, D = determiner, N = noun, NP = noun phrase, P = preposition, PP = prepositional phrase, Pa = particle, S = sentence, t = trace, V = Verb, V' = verb-bar, VP = verb phrase):


AuxP
N
Aux
VP
V
NP
PP
D
N
P
N
They have sent many things to us.
– Phrase structure analysis that allows n-ary branching


AuxP
N
Aux'
Aux
VP
V
VP
NP
V'
D
N
V
PP
P
N
They have sent₁ many things t₁ to us.
– Phrase structure analysis that allows binary branching only

The details in the second diagram here not crucial to the point at hand. This point is that the all branching there is strictly binary, whereas in the first tree diagram ternary branching is present twice, for the AuxP and for the VP. Observe in this regard that strictly binary branching analyses increase the number of (overt) constituents to what is possible. The word combinations *have sent many things to us* and *many things to us* are shown as constituents in the second tree diagram but not in the first. Which of these two analyses is better is again at least in part a matter of what the tests for constituents can reveal.

See also [edit]

- Catena (linguistics)
- Finite verb
- Non-finite verb

Notes [edit]

1. ^ Osborne (2018) provides a detailed and comprehensive discussion of tests for constituents, having surveyed dozens of textbooks on the topic. Osborne's article is available here: Tests for constituents: What they really reveal about the nature of syntactic structure Archived 2018-11-27 at the Wayback Machine. See also Osborne (2019: 2–6, 73–94).
2. ^ This one sentence has been adapted slightly from Radford 1988:91. Radford uses this sentence to introduce and illustrate sentence structure and tests for constituents that identify this structure.
3. ^ Two prominent sources on dependency grammar are Tesnière (1959) and Ágel, et al. (2003/2006).
4. ^ For examples of coordination used as a test for constituent structure, see Baker 1978:269–76; Radford 1981:59–60; Atkinson et al. 1982:172–3; Radford 1988:75–8; Akmajian et al. 1990:152–3; Borsley 1991:25–30; Cowper 1992:34–7; Napoli 1993:159–61; Ouhalla 1994:17; Radford 1997:104–7; Burton–Roberts 1997:66–70; Haegeman and Guéron 1999:27; Fromkin 2000:160–2; Lasnik 2000:11; Lobeck 2000:61–3; Börjars and Burridge 2001:27–31; Huddleston and Pullum 2002:1348–9; van Valin 2001:113–4; Poole 2002:31–2; Adger 2003:125–6; Sag et al. 2003:30; Radford 2004:70–1; Kroeger 2005:91, 218–9; Tallerman 2005:144–6; Haegeman 2006:89–92; Payne 2006:162; Kim and Sells 2008:22; Carnie 2010:115–6, 125; Quirk et al. 2010:46–7; Sobin 2011:31–2; Carnie 2013:99–100; Sportiche et al. 2014:62–8; Müller 2016:10, 16–7
5. ^ The problems with coordination as a test for constituent structure have been pointed out in numerous places in the literature. See for instance Baker 1989:425; McCawley 1998:63; Adger 2003:125; Payne 2006:162; Kim and Sells 2008:22; Carnie 2010:21; Carnie 2013:100; Sportiche et al. 2014:66; Müller 2016:16-7.
6. ^ For examples of pro-form substitution used as a test for constituents, see Allerton 1979:113–4; Radford 1981:63–6; Atkinson et al. 1982:173–4; Radford 1988:78–81, 98–9; Thomas 1993:10–12; Napoli 1993:168; Ouhalla 1994:19; Radford 1997:109; Haegeman and Guéron 1999:46; Fromkin 2000:155–8; Lasnik 2000:9–10; Lobeck 2000:53–7; Börjars and Burridge 2001:24–5; van Valin 2001: 111–2; Poole 2002:29–31; Adger 2003:63; Radford 2004:71; Tallerman 2005:140–2; Haegeman 2006:74–9; Moravcsik 2006:123; Kim and Sells 2008:21–2; Culicover 2009:81; Carnie 2010:19–20; Quirk et al. 2010:75–7; Miller 2011:54–5; Sobin 2011:32; Carnie 2013:98; Denham and Lobeck 2013:262–5; Sportiche et al. 2014:50; Müller 2016:8.
7. ^ For examples of topicalization used as a test for constituents, see Allerton 1979:114; Atkinson et al. 1982:171–2; Radford 1988:95; Borsley 1991:24; Haegeman 1991:27; Napoli 1993:422; Ouhalla 1994:20; Burton–Roberts 1997:17–8; Haegeman and Guéron 1999:46; Fromkin 2000:151; Lasnik 2000:10; Lobeck 2000:47–9; Börjars and Burridge 2001:26; van Valin 2001:112; Poole 2002:32; Adger 2003:65; Sag et al. 2003:33; Radford 2004:72; Kroeger 2005:31; Downing and Locke 2006:10; Haegeman 2006:79; Payne 2006:160; Culicover 2009:84; Quirk et al. 2010:51; Miller 2011:55; Sobin 2011:31; Sportiche et al. 2014:68; Müller 2016:10.
8. ^ For examples of the use of *do-so*-substitution as a test for constituents, see Baker 1978:261–8; Aarts and Aarts 1982:56, Atkinson et al. 1982:174; Borsley 1991:63; Haegeman 1991:79–82; Cowper 1992:31; Napoli 1993:423–5; Burton–Roberts 1997:104–7; Haegeman and Guéron 1999:74; Fromkin 2000:156–7; van Valin 2001:123, 127; Poole 2002:41–3; Tallerman 2005:130–1, 141; Haegeman 2006:75–6; Payne 2006:162; Culicover 2009:81; Carnie 2010:115–6; Quirk et al. 2010:76, 82; Miller 2011:54–5; Sobin 2011:33; Carnie 2013:169–70; Denham and Lobeck 2013:265; Sportiche et al. 2014:61.
9. ^ For examples of *one*-substitution used as a test for constituents, see Baker 1978:327–40, 413–25; Radford 1981:92, 96–100; Aarts and Aarts 1982:57; Haegeman 1991: 26, 88–9; Cowper 1992:26; Napoli 1993:423–5; Burton–Roberts 1997:182–9; McCawley 1998:183; Haegeman and Guéron 1999:75–6; Fromkin 2000: 157–8; van Valin 2001:122, 126, 128, Poole 2002:37–9; Adger 2003:63; Radford 2004:37; Kroeger 2005:97–8; Tallerman 2005:150; Haegeman 2006:109; Carnie 2010:114–5; Quirk et al. 2010:75; Carnie 2013:166–7; Sportiche et al. 2014:52, 57, 60.
10. ^ Concerning the inability of a single constituent structure to simultaneously acknowledge all of the strings that *one*-substitution suggests are constituents, see Cowper 1992:30, Napoli 1993: 425, Burton-Roberts 1997: 187, and Carnie 2013: 190–2.
11. ^ The fact that one-substitution at times suggests that non-string word combinations are constituents is illustrated and discussed by Culicover and Jackendoff 2005:136–9 and Goldberg and Michaelis 2017:4–6.
12. ^ For examples of answer fragments used as a test for constituents, see Brown and Miller 1980:25; Radford 1981:72, 92; Radford 1988:91; Burton–Roberts 1997:15–8; Radford 1997:107; Börjars and Burridge 2001:25; Kroeger 2005:31; Tallerman 2005:125; Downing and Locke 2006:10; Haegeman 2006:82; Moravcsik 2006:123; Herbst and Schüler 2008:6–7; Kim and Sells 2008:20; Carnie 2010:18; Sobin 2011:31; Carnie 2013:98.
13. ^ For examples of clefting used as a test for constituents, see Brown and Miller 1980:25; Radford 1981:109–10; Aarts and Aarts 1982:97–8; Akmajian et al. 1990:150; Borsley 1991:23; Napoli 1993:148; McCawley 1998:64; Haegeman and Guéron 1999:49; Börjars and Burridge 2001:27; Adger 2003:67; Sag et al. 2003:33; Tallerman 2005:127; Downing and Locke 2006:10; Haegeman 2006:85; Kim and Sells 2008:19; Carnie 2013: 98; Sportiche et al. 2014:70.
14. ^ For examples of VP-ellipsis used to test constituent structure, see Radford 1981:67, 1988:101; Napoli 1993:424; Ouhalla 1994:20; Radford 1997:110; McCawley 1998:67; Fromkin 2000:158; Adger 2003:65; Kroeger 2005:82; Tallerman 2005:141; Haegeman 2006:84–5; Payne 2006:163; Culicover 2009:80; Denham and Lobeck 2013:273–4; Sportiche et al. 2014:58–60.
15. ^ For examples of pseudoclefting used as a test for constituents, see Brown and Miller 1980:25; Aarts and Aarts 1982:98; Borsley 1991:24; Napoli 1993:168; McCawley 1998:64; Haegeman and Guéron 1999:50; Kroeger 2005:82; Downing and Locke 2006:10; Haegeman 2006:88; Payne 2006:160; Culicover 2009:89; Miller 2011:56; Carnie 2013:99; Sportiche et al. 2014:71.
16. ^ For examples of passivization used as a test for constituents, see Brown and Miller 1980:25; Borsley 1991:24; Thomas 1993:10; Lobeck 2000:49–50; Downing and Locke 2006:10; Carnie 2010:21; Sobin 2011:30; Carnie 2013:99; Denham and Lobeck 2013:277.
17. ^ For examples of omission used as a test for constituents, see Allerton 1979: 113–9; Aarts and Aarts 1982: 60–1, 65–7; Burton–Roberts 1997: 14–5; Börjars and Burridge 2001: 33–4; Payne 2006: 163–5; Carnie 2010: 19; Hudson 2010: 147; Quirk et al. 2010: 41, 51, 61; Miller 2011: 54; Sobin 2011: 33).
18. ^ For examples of intrusion used as a test for constituents, see Radford 1981:60–2; 1988:93; McCawley 1998:68–70; Fromkin 2000:147–51; Börjars and Burridge 2001:34; Huddleston and Pullum 2002:21; Moravcsik 2006:123; Payne 2006:162.
19. ^ For examples of wh-fronting used as a test for constituents, see Radford 1981:108; Haegeman 1991:28; Haegeman and Guéron 1999:46–7; Lobeck 2000:57–9; Payne 2006:160; Culicover 2009:90–1; Denham and Lobeck 2013:279–81; Sportiche et al. 2014:58–60; Müller 2016:9.
20. ^ For examples of the general substitution test, see Allerton 1979: 113; Brown and Miller 1980: 38; Aarts and Aarts 1982: 11; Radford 1988: 89–91; Moravcsik 2006: 123–4; Culicover 2009: 37; Quirk et al. 2010: 41; Müller 2016: 7–8.
21. ^ For examples of RNR used as test for constituents, see Radford 1988: 77–8, 97; Radford 1997: 106; McCawley 1998: 60–1; Haegeman and Guéron 1999: 52, 77; Sportiche et al. 2014: 67–8.
22. ^ For a comparison of these two competing views of constituent structure, see Osborne (2019:73-94).

References [edit]

- Adger, D. 2003. Core syntax: A minimalist approach. Oxford, UK: Oxford University Press.
- Ágel, V., L. Eichinger, H.-W. Eroms, P. Hellwig, H. Heringer, and H. Lobin (eds.) 2003/6. Dependency and valency: An international handbook of contemporary research. Berlin: Walter de Gruyter.
- Akmajian, A., R. Demers, A. Farmer and R. Harnish. 2001. Linguistics: An introduction to language and communication, 5th edn. Cambridge: MIT Press.
- Allerton, D. 1979. Essentials of grammatical theory: A consensus view of syntax and morphology. London: Routledge and Kegan Paul.
- Aarts, Flor and Jan Aarts. 1982. English syntactic structures: Functions & categories in sentence analysis. Oxford, UK: Pergamon Press and Bohn: Scheltema & Holkema.
- Atkinson, M., D. Kilby, and Iggy Roca. 1982. Foundations of general linguistics, second edition. London: Unwin Hyman.
- Baker, C. L. 1978. Introduction to generative transformational grammar. Englewood Cliffs, NJ: Prentice-Hall.
- Baker, C. L. 1988. English syntax. Cambridge, MA: The MIT Press.
- Börjars, K. and K. Burridge. 2001. Introducing English grammar. London: Arnold.
- Borsley, R. 1991. Syntactic theory: A unified approach. London: Edward Arnold.
- Brinker, K. 1972. Konstituentengrammatik und operationale Satzgliedanalyse: Methodenkritische Untersuchungen zur Syntax des einfachen deutschen Satzes. Frankfurt a. M.: Athenäum.
- Brown, K. and J. Miller 1980. Syntax: A linguistic introduction to sentence structure. London: Hutchinson.
- Burton-Roberts, N. 1997. Analysing sentences: An introduction to English syntax. 2nd Edition. Longman.
- Carnie, A. 2002. Syntax: A generative introduction. Oxford: Blackwell.
- Carnie, A. 2010. Constituent Structure. Oxford: Oxford University Press.
- Carnie, A. 2013. Syntax: A generative introduction. Malden, MA: Wiley-Blackwell.
- Cowper, E. 1992. A concise introduction to syntactic theory: The government-binding approach. Chicago: The University of Chicago Press.
- Culicover, P. 2009. Natural language syntax. Oxford, UK: Oxford University Press.
- Culicover, P. and . Jackendoff. 2005. Simpler syntax. New York: Oxford University
- Dalrymple, M. 2001. Lexical functional grammar. Syntax and semantics 34. San Diego: Academic Press.
- Denham, K. and A. Lobeck. 2013. Linguistics for everyone: An introduction. Boston: Wadsworth Cengage Learning.
- Downing, A. and P. Locke. 2006. English grammar: A university course, 2nd edition. London: Routledge.
- Fromkin, V. A. (ed.). 2000. An introduction to linguistic theory. Malden, MA: Blackwell Publishing.
- Goldberg, A. and L. Michaelis. 2017. One among many: Anaphoric one and its relationship with numeral one. Cognitive Science, 41.S2: 233–258.
- Haegeman, L. 1991. Introduction to Government and Binding Theory. Oxford, UK: Blackwell.
- Haegeman, L. 2006. Thinking syntactically: A guide to argumentation and analysis. Malden, MA: Blackwell.
- Haegeman, L. and J. Guéron 1999. English grammar: A generative perspective. Oxford: Basil Blackwell.
- Herbst, T. and S. Schüller. 2008. Introduction to syntactic analysis: A valency approach. Tübingen: Narr.
- Huddleston, R. and G. Pullum. 2002. The Cambridge grammar of the English language. Cambridge, UK: Cambridge University Press.
- Hudson, R. 2010. An introduction to Word Grammar. Cambridge, UK: Cambridge University Press.
- Jacobson, P. 1996. Constituent structure. In Concise encyclopedia of syntactic theories. Cambridge: Pergamon.
- Kim, J. and P. Sells. 2008. English syntax: An introduction. Stanford: CSLI Publications.
- Kroeger, P. 2005. Analyzing grammar: An introduction. Cambridge University Press.
- Lasnik, H. 2000. Syntactic structures revisited: Contemporary lectures on classic transformational theory. Cambridge: MIT Press.
- Lobeck, A. 2000. Discovering grammar: An introduction to English sentence structure. New York: Oxford University Press.
- McCawley, J. 1998. The syntactic phenomena of English, 2nd edn. Chicago: University of Chicago Press.
- Miller, J. 2011. A critical introduction to syntax. London: Continuum.
- Moravcsik, E. 2006. An introduction to syntax: Fundamentals of syntactic analysis. London: Continuum.
- Müller, Stefan. 2016. Grammatical theory: From transformational grammar to constraint-based approaches (Textbooks in Language Sciences 1). Berlin: Language Science Press.
- Napoli, D. 1993. Syntax: Theory and problems. New York: Oxford University Press.
- Nerbonne, J. 1994. Partial verb phrases and spurious ambiguities. In: J. Nerbonne, K. Netter and C. Pollard (eds.), German in Head-Driven Phrase Structure Grammar, CSLI Lecture Notes Number 46. 109–150. Stanford: CSLI Publications.
- Osborne, Timothy. 2018. Tests for constituents: What they really reveal about the nature of syntactic structure. Language Under Discussion 5, 1, 1–41.
- Osborne, T. 2019. A Dependency Grammar of English: An Introduction and Beyond. Amsterdam: John Benjamins. https://doi.org/10.1075/z.224
- Ouhalla, J. 1994. Introducing transformational grammar: From rules to principles and parameters. Oxford: Oxford University Press.
- Payne, T. 2006. Exploring language structure: A student's guide. New York: Cambridge University Press.
- Poole, G. 2002. Syntactic theory. New York: Palgrave.
- Quirk, R., S. Greenbaum, G. Leech, and J. Svartvik. 2010. A com-prehensive grammar of the English language. Dorling Kindersley: Pearson.
- Radford, A. 1981. Transformational syntax: A student's guide to Chomsky's Ex-tended Standard Theory. Cambridge, UK: Cambridge University Press.
- Radford, A. 1988. Transformational grammar: A first course. Cambridge, UK: Cambridge University Press.
- Radford, A. 1997. Syntactic theory and the structure of English: A minimalist approach. Cambridge, UK: Cambridge University Press.
- Radford, A. 2004. English syntax: An introduction. Cambridge, UK: Cambridge University Press.
- Sag, I., T. Wasow, and E. Bender. 2003. Syntactic theory: A formal introduction, 2nd edition. Stanford: CSLI Publications.
- Santorini, B. and A. Kroch 2000. The syntax of natural language: An online introduction using the trees program. Available at (accessed on March 14, 2011): http://www.ling.upenn.edu/~beatrice/syntax-textbook/00/index.html Archived 2016-12-28 at the Wayback Machine.
- Sobin, N. 2011. Syntactic analysis: The basics. Malden, MA: Wiley-Blackwell.
- Sportiche, D., H. Koopman, and Edward Stabler. 2014. An introduction to syntactic analysis. Malden, MA: Wiley-Blackwell.
- Tallerman, M. 2005. Understanding syntax. London: Arnold.
- Tesnière, L. 1959. Éléments de syntaxe structurale. Paris: Klincksieck.
- van Valin, R. 2001. An introduction to syntax. Cambridge, UK: Cambridge University Press.

Categories: Generative syntax | Syntax

This page was last edited on 16 July 2025, at 15:26 (UTC).

Text is available under the Creative Commons Attribution-ShareAlike 4.0 License; additional terms may apply. By using this site, you agree to the Terms of Use and Privacy Policy. Wikipedia® is a registered trademark of the Wikimedia Foundation, Inc., a non-profit organization.